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The Routine: Enhancing Special Needs Education Through AI-Personalized Embodied Learning in Virtual Environments

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***Abstract*—"The Routine" is a virtual learning environment (VLE) designed for children with special needs, primarily those with autism. By integrating embodied learning theory, which combines physical movement with cognitive development it aims to teach life skills and promote independence. Unlike traditional approaches, "The Routine" employs artificial intelligence to personalize learning experiences and dynamically adjust to each learner's needs. Through controlled play, intelligent educational agents enhance behavioral, social, and cognitive abilities in real-time. The system incorporates wearable electronics and sensors to track each child's physical and emotional states, adapting accordingly to provide a tailored experience. Future developments will focus on improving spatial learning through the integration of virtual and mixed-reality technologies, as well as enhancing prototypes with adjustable hardware to increase accessibility. As an ongoing project in educational technology, "The Routine" emphasizes adaptive, interactive, and multimodal learning methodologies, underscoring the revolutionary potential of VLEs in special education.**

Keywords— VLE, Embodied Learning, Autism Therapy, Wearable Technologies, Narrative Game Based-Design

I. Introduction

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition that affects how individuals perceive and interact with the world, often influencing communication, social skills, and behaviors in ways that vary widely across the spectrum [1]. ASD presents unique challenges in educational settings, particularly for young children, as traditional classroom-based instruction lacks the flexibility and adaptability needed to address their diverse sensory and developmental needs. Furthermore, conventional therapies for ASD like Applied Behavioral Analysis (ABA) and Pivotal Response Treatment (PRT) rely on physical, analog toys which, while useful, do not adapt dynamically to each child's needs nor provide interactive stimuli essential for engagement [2].

Virtual Learning Environments (VLEs) are interactive digital platforms that utilize technologies such as simulations, gamification, and adaptive tools to create dynamic and personalized learning experiences, providing flexible access to resources and communication channels beyond traditional classroom settings [3]. They represent a transformative opportunity in special needs education by integrating immersive technologies such as virtual and mixed reality, along with artificial intelligence (AI), to deliver highly personalized and interactive experiences for learners. This paper introduces "The Routine," a gamified VLE designed to integrate physical movement with cognitive development, fostering both mental and physical engagement for children with ASD. This immersive game extends interaction beyond traditional screen-based devices, dynamically adapts educational content in real-time, and utilizes an AI-powered learning companion to provide immediate feedback through a reward system, enhancing interaction and optimizing learning outcomes [4].

II. Theoretical and Empirical Foundations

VLEs are designed information spaces that enhance learning through interactive tools like videos, quizzes, and forums, facilitating adaptive and tailored interventions suitable for children with ASD [5], [6]. Further enhancing VLEs, immersive technologies such as virtual and mixed reality create engaging environments that significantly improve emotional regulation and learning outcomes. These technologies provide dynamic educational tools that allow for safe development of cognitive, social, and emotional skills [3]. Complementing this, AI tailors learning experiences in real-time, using data from user interactions to adjust educational content and strategies, enriching the learning process through personalized feedback and simulated social exchanges [7], [8].

Gamification elements, underpinned by well-established learning theories such as the Objectives, Activity, Accomplishment, and Affect (OA3) and the Player Engagement Process (PEP), motivate continued engagement and skill development [9]. Studies highlight the effectiveness of such approaches in special needs education, promoting persistent interaction and fostering essential life skills [10], [11]. Incorporating physical movement into this educational framework not only enhances physical well-being but also improves cognitive functions such as memory and problem-solving, crucial for holistic development in children with special needs [12], [13], [14].

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III. DESIGN

A. Game Overview and Design Philosophy

"The Routine" is a serious game developed to make learning interactive, engaging, and adaptive for children with special needs, particularly ASD. Rooted in embodied and situated learning theories, the game employs body movement to simulate real-life experiences and teach essential life skills. Through the integration of VLE, immersive display, artificial intelligence (AI), and gamification, the game provides a holistic and personalized learning experience tailored to the diverse sensory and developmental needs of its players. Currently in a low-fidelity development phase, "The Routine" demonstrates basic interactions and design elements, with plans to integrate mixed reality (MR) technologies in subsequent iterations to create a fully immersive environment.

Informed by embodied learning theory, which integrates physical actions with cognitive development [9], "The Routine" specifically aims to enhance cognitive and social behaviors through engaging game designs by reducing the need for the permanent interventions of occupational therapists while offering a customizable learning system tailored to each learner's interests and needs [15]. The flexibility of the VLE allows for modifications at different levels to meet the specific challenges of each child, distinguishing it from conventional educational systems designed for neurotypical students [16], [17]. This adaptability is crucial for effectively addressing the varied cognitive and social needs of children with special needs.

B. Characters and Interactions

In "The Routine," interaction between players and the virtual environment is facilitated through two main characters: the player's avatar and a conversational AI learning companion embodied as an anthropomorphized rabbit , whom we will refer to as "Mochi". The player's avatar allows for a high degree of personalization, enhancing the player's sense of presence and navigation within the virtual world (Fig.1). This personalization not only engages players more deeply but also assists them in navigating the complexities of identity, which can be particularly challenging for children with special needs who may face difficulties with social interaction and maintaining attention [18], [19], [20]. Mochi enhances this interaction by providing customized feedback and support, thereby impacting effective learning [21], [22], [23]. It adapts the difficulty of tasks based on the player's prior knowledge and progress, aiding in decision-making, problem-solving, and various stages within the game.



Fig. 1. The profile of the user, including user's avatar, showing points, sound and visual cues, customization, and option for collaborative mode, made with ChatGPT

"The Routine" features gameplay that mimics real-life experiences, encouraging players to engage in daily routines such as dressing, eating, and personal hygiene through sensor-recognized physical gestures through interactive gameplay. For example, simulating tooth brushing involves players mimicking the actual motion, which is captured and processed in real-time for feedback. The game is structured into levels that represent different parts of a typical day, from morning routines to playground interactions, promoting skill development and socialization (Fig. 2). Gamification elements like points, badges, and rewards are integrated to motivate players and reinforce positive behaviors, ensuring a compelling and educational experience (Fig. 1).

C. Reward System

"The Routine" incorporates a reward system, intricately linked to the dynamics between the player's avatar, Mochi, and the player's performance, thereby enriching the learning experience. Central to this system are the tactile and motor skills, which serve as core motivators. This system encourages players, particularly those with ASD, to engage physically with the game. For example, players might simulate brushing their teeth by executing precise up and down and circular motions, or they might open closets by performing a pulling motion. These actions are recorded in real-time through motion capture and gesture recognition. The implementation of these actions not only engages the player physically but also aids in reinforcing the muscle memory associated with daily tasks [19]. Successful completion of tasks will trigger multi-sensory feedback, such as a sparkling clean tooth animation, auditory applause as well as encouraging words from Mochi to reinforce learning achievements. Additional auditory cues, such as the sound of water running or a door creaking, also enhance realism, making the learning experience more vivid and engaging [24].

Dynamic personalization is a key feature of "The Routine" the game adjusts the game's difficulty in real time based on the player's performance and learning progress. This includes simplifying tasks if the player struggles or offering more complex scenarios as their skills develop. The system ensures that the learning experience is optimized to each individual's capabilities and learning goals, thereby maximizing educational outcomes. Future development of the game will incorporate virtual social interactions. As players progress, they will be introduced to collaborative games designed to promote effective social exchanges and emotional development. Mochi will facilitate such interactions by introducing and pairing the player with AI-powered non-player characters (NPC) or other real players at similar or higher developmental stages, actively managing the game dynamics to ensure all participants are engaged and contributing. For instance, in games designed to enhance turn-taking skills, Mochi might prompt players when it is their turn and provide encouragement or hints to help them perform their part effectively. This element of co-play, supported by Mochi's intelligent interaction, enriches the players' experience, making the learning journey immersive and contextual. Each task within these collaborative settings is meticulously crafted to align with the educational goals of "The Routine," ensuring that the development of social skills is integrated with core learning objectives [22], [25].

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Fig. 2. Low-fidelity prototype of The Routine on Figma; The Customization, Characters, Seasons, Main Map, Location, Rooms, Bathroom level, Brush level (Selection through click, touch and hand gestures with the help of a therapist)

IV. LEARNING GOALS

The core of the game is to foster independence through structured engagement with daily routines such as dressing, eating, and maintaining personal hygiene. By navigating through multiple game phases and levels, children master these routines by engaging physically, thereby enhancing their ability to function independently in everyday tasks

The game also places a strong emphasis on social interaction and behavioral skills. Through physical interaction with the game environment, children practice and develop social behaviors and communication skills within simulated real-world settings like schools or playgrounds [26], [27]. These interactions are designed to mirror real-life social scenarios, making them invaluable for developing negotiation and cooperative skills as part of situated [28], [29]. Enhanced social interaction is further supported by the game's pedagogical agents, who are customized to match each child's unique needs and preferences, guiding them through tasks and providing real-time feedback and support [23]. Additionally, "The Routine" is deeply committed to improving behavioral skills through problem-solving and decision-making tasks that require thoughtful and intentional movements. These actions are supported by a reward system that provides positive reinforcement through auditory and visual cues, fostering an environment where correct responses lead to rewards, thus promoting appropriate behavior [30].

A. Social and Behavioral Interaction

The game progresses through levels that replicate various social contexts—from simple home interactions to complex activities in school settings—helping children develop crucial skills such as decision-making, task completion, and cooperation. This structured approach not only enhances their ability to engage within a community but also reinforces appropriate behaviors through rewards for positive social exchanges (see Table I).

B. Body Movement

"The Routine" integrates body movement as a fundamental element in the learning process, effectively using physical actions to facilitate educational outcomes and independence in daily tasks. Each movement within the game mimics real-life activities, such as adjusting the time of day by moving hands in a circular motion, akin to setting a clock, or navigating different seasons through specific gestures. This approach not only makes learning concepts like time more tangible but also significantly enhances motor skills, spatial awareness, and sensory integration [28].

The game further exemplifies embodied learning principles by making physical movement central to the learning process, a concept also discussed by [31], [32]. By translating physical actions into responsive game mechanics, the game ensures that children not only comprehend but can perform daily tasks independently, thereby remaining mentally and physically engaged. This dual engagement supports overall cognitive development and health, effectively showcasing how embodied learning can be leveraged within virtual environments to enrich educational experiences for children, particularly those with special needs (Fig. 3).

C. User Experience and Levels

The Routine" provides a highly intuitive and engaging user experience by creating a realistic immersive environment where children interact with virtual objects through hand gesture and body movements. This design minimizes the need for handheld controllers and screen devices, leveraging advanced body-tracking and gesture-recognition technologies to promote full-body movement. Such interactions are not only crucial for motor skill development but also ensure that players remain physically active, countering the sedentary tendencies of traditional screen-based learning. From the onset, players personalize their experience by setting up profiles, selecting customizable avatars, and choosing initial difficulty levels, which can be adjusted as they progress to continually match their developing skills. This customization enhances user agency, a key factor in effective learning, as it allows learners to shape their learning environment to better suit their needs and preferences [33].

As players navigate through "The Routine," they encounter levels that represent various parts of a typical day, from morning routines at home to activities at school or visits to a relative's house. Each level is designed to be accessible and engaging, equipped with assistive features like voice prompts and visual icons to help guide players through tasks [30]. These tasks are more than just game stages; they are integral parts of a therapeutic process aimed at practicing and reinforcing daily living skills. "The Routine" enhances player immersion and agency through interaction without handheld devices and detailed environmental cues. This design fosters a more authentic virtual experience, improving educational outcomes by boosting enjoyment, skill development, and task performance [25], [34], [35].

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TABLE I LEARNING GOALS AND SUB-GOALS OF THE ROUTINE

Main Goal	Sub Goal
Enhancing Independence	Mastery of daily routines such as dressing, eating, and personal hygiene through sequential game levels.
	Developing the ability to initiate and complete tasks without direct supervision, using the game's feedback system to guide progress.
Promoting Social Interaction	Encouraging interaction with virtual peers and environments, simulated by the game to mirror real-life social settings like schools or playgrounds.
	Facilitating communication skills through structured interactions within the game, such as negotiating, sharing, and collaborating on tasks.
Improving Behavioral Skills	Teaching appropriate responses to everyday situations through simulated scenarios that require problem-solving and decision-making.
	Reducing behavioral challenges by providing consistent, positive reinforcement for appropriate behaviors within the game environment.

V. IMPLEMENTATION

A. Gameplay

Players begin by customizing their environment and avatars, adjusting settings such as color, brightness, and sound to suit their sensory preferences. As they progress through levels—from learning about time to performing daily routines—players interact with the game using physical movements and decision-making, guided by real-time visual and auditory cues. These interactions, which require fine motor skills and coordination, are captured through embedded cameras and sensors, enabling a seamless integration of educational goals with engaging gameplay. Players navigate tasks by selecting locations on a map, completing challenges that reflect real-life scenarios, and receiving immediate feedback to reinforce learning (see Table II).

The first low-fidelity prototype of the game, developed in Figma, demonstrates the customizable settings, diverse environments, game levels, interactions, characters, and avatars in a narrative format (Fig.2). The next step involves transitioning to a high-fidelity prototype designed specifically for Apple Vision Pro, which will integrate advanced wearable technologies, gesture tracking, and AI-enhanced interactions.

B. Technology

"The Routine" leverages a robust technological framework designed to optimize the interactive learning experience for children with special needs. This includes AI-powered pedagogical agents for real-time guidance, gesture recognition for natural interactions, and tools for tracking emotions and physical activity (Fig.4). As we move into the next phase of development, we are building the high-fidelity prototype for Apple Vision Pro while also envisioning future implementations with complementary technologies such as Magic Mirror systems and room-scale projection.

1) Apple Vision Pro for the Next Phase: The next development phase focuses on leveraging Apple Vision Pro's advanced spatial computing capabilities. Unlike traditional VR headsets, Apple Vision Pro allows for seamless blending of digital elements with the physical world. Its gesture-based interaction and eye-tracking features eliminate the need for handheld controllers, making it particularly suited for children with special needs. The device's app-based development platform enables easy customization and accessibility features such as voice commands and adaptive sensory settings. This ensures that "The Routine" is inclusive for a diverse range of abilities and creates a natural, intuitive learning environment. Apple Vision Pro also minimizes the complexity of setup, making it practical for educational environments that require portability and ease of use.

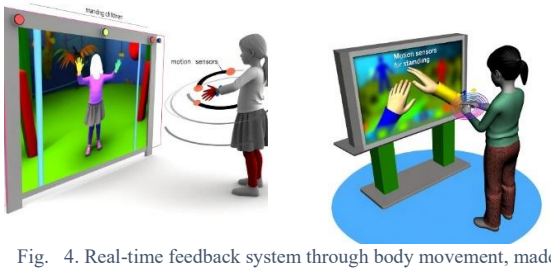


Fig. 4. Real-time feedback system through body movement, made with ChatGPT



Fig. 3. Social interaction within the story (Cooperative and Collaborative), made with ChatGPT

2) Future Technology Exploration: While Apple Vision Pro provides the primary platform for the next phase, we are exploring additional technologies to further enrich the learning experience:

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TABLE II EXPERIENCE DESIGN OF EACH LEVEL AND TASK

Levels	Game Tasks	Interactions/ Experience	Technology
1- Initial Customization	Customizing game environment, personal profile settings.	Players adjust game settings like sound and color preferences.	The user interface allows for easy adjustments.
2- Avatar Creation	Creating their own avatar as their virtual representative in the game.	Creation of a personalized avatar that represents the player.	Advanced customization software for detailed avatar creation.
3- Pedagogical Agent Customization	Customizing an individual pedagogical agent based on player's interests and limitations.	Customization of a pedagogical agent tailored to the player's needs.	AI technology adapts the agent's behavior based on player interactions.
4- Understanding Time	Becoming familiar with the concept of time by selecting seasons, months, days, and changing the time.	Players use gestures to select from a menu of seasons, then drill down to month and day; moving hands in a circular motion to change the time of day, simulating a clock movement.	Gesture detection translates movements into game responses.
5- Profile Stage	Viewing and managing their profile.	Players view their avatar, pedagogical agent, rewards, collaborative mode options, and the selected time. Assistive voice and visual icons aid in understanding their position.	Interactive elements and feedback mechanisms provide user guidance.
6- Map Level	Selecting a location to perform daily life skills.	Interactive map where players choose a destination like Home, School, Playground, or a relative's house to engage in activities related to each place.	Dynamic map interface with touch interaction for location selection.
7- Morning Routine at Home	Waking up, brushing teeth, getting dressed.	Engaging in routine morning tasks such as brushing teeth and dressing through simulated physical movements.	Gesture detection technology monitors and responds to physical movements.
8- School Day	Attending classes, interacting with peers, learning new topics.	Body movements to simulate walking between classes, hand gestures for writing or interacting.	AI-driven avatars provide feedback and guide learning activities.
9- Playground Interaction	Playing games, interacting with other children.	Full-body movement games that require jumping, running on the spot.	Real-time feedback on movements and interactions enhances engagement.
10- Visit to a Relative's House	Social interactions, participating in a family meal.	Gestures for eating, conversing with relatives through pre-set dialogues triggered by specific movements.	Speech recognition, response, evaluate and support social interaction effectiveness.

- **Magic Mirror System:** A magic mirror setup, using a smart screen or projector, overlays virtual objects onto the child's real-world reflection. This system creates a natural and intuitive interface, enabling children to
- interact with virtual objects like clothes or tools without needing external devices. This approach is particularly beneficial for tasks like dressing, where seeing oneself in the virtual mirror aids in skill acquisition and engagement.
- **Room-Scale Projection**: Room-scale projection systems allow virtual elements to be mapped onto physical spaces. For example, virtual footprints could guide a child through a room or clocks projected on walls could teach time-management skills. These systems foster a sense of presence and make learning more tangible while reducing the reliance on wearable devices.
- **Wearable Sensors and AI Tracking:** Wearable technologies such as wristbands or smartwatches can track physiological metrics like heart rate or stress levels, enabling the game to adapt dynamically to a child's emotional and physical state. For example, if stress is detected during a task, the game could lower sensory input or provide calming feedback via the Pedagogy Agent.
- **Facial Recognition and Emotion Detection:** Embedded cameras could analyze facial expressions

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to provide real-time adjustments to gameplay. If frustration is detected, the AI could simplify the task or offer encouragement, fostering a more supportive learning environment.

3) Balancing Feasibility and Innovation: While these additional technologies offer exciting possibilities for future implementation, their integration depends on overcoming current technological and logistical limitations. For example, room-scale projection systems may require dedicated spaces and additional hardware, which might limit accessibility for certain users. Similarly, wearable sensors, though promising, need to balance affordability and ease of use to ensure they remain practical for widespread application.

4) A Forward-Looking Strategy: By focusing on Apple Vision Pro for the next phase of development, we aim to build a foundation that maximizes accessibility, adaptability, and immersion. At the same time, exploring complementary technologies such as Magic Mirror systems, room-scale projection, and wearable sensors ensures that "The Routine" can evolve into a more comprehensive educational platform. These future developments will enable the game to better address the diverse and complex needs of children with special needs, paving the way for a truly inclusive and innovative learning environment.

VI. Conclusion

"The Routine" represents a significant advancement in special needs education through its innovative use of Virtual Learning Environments (VLEs), artificial intelligence (AI), and immersive technologies tailored for children with Autism Spectrum Disorder (ASD). By integrating dynamic, AI-driven pedagogy within a virtual setting, this platform personalizes learning experiences to each child's unique needs, thereby optimizing educational outcomes. Key contributions of this research include enhanced engagement and learning through interactive gameplay that encourages physical movement and cognitive development. Moreover, "The Routine" utilizes a mixed-reality approach to provide naturalistic learning environments that improve the user experience and facilitate motor skill development without the constraints of traditional screen-based learning. However, the project is still in its developmental phase, with ongoing efforts to refine technology integration and expand content accessibility. Future work will focus on enhancing spatial learning capabilities, improving interface adaptability for broader accessibility, and conducting empirical studies to rigorously evaluate the system's effectiveness in real-world educational settings. By continuously advancing the technological framework and content adaptability, "The Routine" aims to provide an even more inclusive and effective educational tool, shaping the future of learning for children with special needs.

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